JOHNCOOL PHARMACY LIMITED

Complete Pharmacy Management & Inventory Software Requirements

Prepared as a functional and technical requirements document for software development

Objective: Develop a scalable pharmacy management platform covering inventory, purchasing, POS, batch/expiry control, staff accountability, profitability, warehouse and multi-branch management, remote monitoring and future e-commerce integration.

1. System Purpose

- Complete pharmacy management system for Johncool Pharmacy's current operation and future expansion.
- Modules must cover inventory, purchasing, suppliers, POS/sales, batch and expiry management, staff, customers, finance, warehouse, branches, remote monitoring and online sales.

2. Dashboard

- Today's sales, transactions, gross profit, cash/POS/transfer/credit sales, stock value, product count, low/out-of-stock items, near-expiry/expired items, customer credit, supplier balances, purchases, expenses, estimated net profit, top/slow-moving products and staff performance.
- Filters: today, yesterday, week, month, custom period, branch, staff and category.

3. Product Master Database

- Product name, generic/brand name, category/sub-category, dosage form, strength, pack size, unit, manufacturer, supplier, description, barcode/SKU, OTC/prescription classification, restricted-medicine classification where applicable, image, status, cost price, selling price, wholesale price, minimum price, reorder level, maximum stock and tax/VAT status.
- Support carton, pack, bottle, strip, sachet, piece, vial and ampoule units with correct stock conversion.

4. Batch Management

- Each batch must be separately identifiable by batch number, manufacturing date, expiry date, received quantity, sold quantity, balance, cost, selling price, supplier and purchase invoice.
- Multiple batches of the same product must be supported without incorrectly merging their records.

5. Expiry Management

- Automatic monitoring for expired, 30-, 60-, 90-day, 6-month and 12-month expiry windows. Configurable alerts and branch/batch-specific reports.

6. FEFO Stock Control

- First-Expiry-First-Out should recommend/use the earliest-expiring batch first. Authorized overrides must be logged in the audit trail.

7. Purchase Management

- Purchase orders, quotations, invoices, receiving, partial/full delivery, purchase returns, cash/credit purchases, discounts, additional costs, cost-price updates, batch and expiry entry.

8. Supplier Management

- Supplier profile, contacts, products supplied, purchase history, balances, payments, credit terms, invoices and supplier performance.

9. Sales / POS

- Fast POS with barcode scanning, product/generic/brand/SKU/category search, quantity changes, discounts, permission-controlled price override, customer selection, batch/FEFO selection, prescription warning, suspended transactions, returns/refunds and receipt reprinting.

10. Payment Methods

- Cash, POS/card, bank transfer, credit and split/multiple payments in one transaction.

11. Receipts

- Receipt must show Johncool Pharmacy identity, branch, address/contact, transaction number, date/time, cashier, items, quantities, prices, discounts, totals, payment method and change/balance. Printable and optionally digital.

12. Barcode System

- Support existing barcodes, barcode generation/printing, scanning, internal SKUs and multiple barcodes per product.

13. Sales Returns

- Return against original transaction, batch selection, reason, appropriate stock restoration or damaged/expired classification, with full audit trail.

14. Stock Management

- Real-time stock movement for opening stock, purchases, sales, returns, damaged/expired stock, adjustments, transfers, samples and internal use. Every movement must record date/time, user, product, quantity, reason and before/after balance.

15. Stocktaking

- Physical stock counts, barcode-assisted counting, blind counts, variance reports, approval of adjustments and stocktake history.

16. Low-Stock Management

- Minimum/reorder/maximum levels, low/out-of-stock/overstock alerts and suggested purchase orders based on stock and sales history.

17. Warehouse Management

- Central warehouse, branch stock, receiving, issuing, transfer requests/approvals, transfer history and stock-in-transit.

18. Multi-Branch Management

- Separate stock, sales, expenses, staff and cashier records per branch, with head-office consolidated and branch-specific reporting.

19. Staff Management

- Individual accounts and role-based permissions for Super Admin, CEO/Owner, Branch Manager, Pharmacist, Superintendent Pharmacist, Cashier, Inventory Officer, Sales Staff, Accountant and Warehouse Officer.

20. Staff Accountability

- Record logins, sales, discounts, price changes, returns, voids, stock adjustments, purchases and sensitive edits. Every important action must identify the user.

21. Cashier Reconciliation

- End-of-shift/day closing showing expected/actual cash, variance, POS, transfer, credit, refunds, discounts and sales totals. Manager approval should be available.

22. Customer Management

- Customer profiles, purchase history, credit balance and payment history. Prescription/medication information only where lawfully collected and protected.

23. Credit Sales

- Credit accounts, limits, due dates, part-payments, balances, payment history, statements and overdue alerts. Authorized approval required.

24. Expense Management

- Record rent, electricity, internet, transport, salaries, repairs, fuel, marketing, bank/POS charges and other expenses by branch/category.

25. Profit & Loss

- Sales revenue, cost of goods sold, gross profit, operating expenses and estimated net profit, with reports by product, category, staff, period and branch.

26. Product Performance

- Fast-moving, slow-moving, most profitable, low-margin and dead-stock reports with configurable periods.

27. Reporting

- Daily/weekly/monthly/annual sales, profit, purchases, suppliers, stock valuation/movement, expiry, batches, low/out-of-stock, damaged/expired stock, returns, customer credit, supplier debt, staff, branch, cashier and expenses.
- Reports must export to Excel, CSV and PDF.

28. CEO / Remote Monitoring

- Remote phone/computer dashboard for sales, branch performance, profit, stock value, low/expiry stock, staff activity, cashier performance, debts and expenses.
- Optional alerts for large discounts/refunds, significant stock adjustments, low stock, expiry, suspicious activity, cash shortages and sync failures.

29. Online Pharmacy / E-commerce

- Future-ready product catalogue, availability, online orders, customer details, order status, delivery, payment integration, branch fulfilment and synchronized stock.

30. Prescription Management

- Where legally appropriate: prescription attachment, prescription number, prescriber, pharmacist verification, dispensed items, quantity and date. Appropriate prescription-required warnings.

31. Controlled/Restricted Medicines

- Configurable restricted access, authorized dispensing, special records, batch/quantity tracking and detailed audit trail in line with applicable Nigerian requirements.

32. Inventory Import

- Excel/CSV import for product name, generic/brand, strength/form, pack size, barcode, batch, expiry, quantity, cost, selling price and supplier. Validate before publishing.

33. Data Export

- Johncool Pharmacy must be able to export products, inventory, batches, expiry, purchases, sales, customers, suppliers, staff and reports in standard formats.

34. Backup & Data Security

- Automatic/cloud backups, recovery, authentication, strong passwords, role-based access, session management, encryption where appropriate and disaster recovery.

35. Internet / Offline Mode

- POS should continue during temporary internet outages and synchronize safely when connectivity returns, without duplicate transactions.

36. Notifications

- Dashboard and optional SMS/email/WhatsApp notifications for low stock, expiry, expired products, large discounts, adjustments, sync failures, cash variance and overdue credit.

37. Security & Audit Trail

- No important transaction should simply disappear. Maintain auditable records for sales, purchases, returns, discounts, price changes, adjustments, user changes, voids and login activity. Prefer voiding/marking over permanent deletion.

38. System Administration

- Configure branches, warehouses, staff, roles, categories, payment methods, tax settings, receipt design, printer/barcode settings and expiry/reorder alerts.

39. Hardware Compatibility

- Must work with standard Windows POS computer, touchscreen where available, keyboard, mouse, 80mm thermal printer, barcode scanner, cash drawer and label/barcode printer. Avoid proprietary hardware dependence.

40. Software & Data Ownership

- Johncool Pharmacy owns its inventory, sales, purchases, customer, supplier, financial and business data. The provider must not lock the business out of its data. Provide a clear export procedure if service ends.

41. Technical Support

- Installation, setup, staff training, remote support, bug fixes, updates, recovery support and documentation. Agree a defined support response time.

42. Development & Testing

- Use a test environment before deployment. Test product import, purchases, batches, expiry, sales, barcode scanning, stock deduction, returns, stocktake, expiry reports, profit, permissions, transfers, cashier closing, reporting, backup and export before full migration.

43. Future Development

- Architecture should support mobile app, loyalty, WhatsApp ordering, online payment, delivery tracking, accounting/payroll integration, AI forecasting, automatic reordering, supplier marketplace, multiple warehouses, BI and franchise management.

44. Priority Features

Priority	Mandatory/Important Features
Tier 1 — MUST HAVE	Product database; batch; expiry; FEFO; stock; purchasing; POS; barcode; automatic stock deduction; stocktaking; low-stock ale
Tier 2 — VERY IMPORTANT	Warehouse; multi-branch; stock transfers; customer credit; expenses; cashier reconciliation; expiry alerts; slow/dead-stock repo
Tier 3 — FUTURE	Mobile app; WhatsApp ordering; loyalty; delivery; AI forecasting; accounting/payroll integration.

45. Acceptance / Success Criteria

✓ Management can know exact current stock.

✓ Management can identify each batch and expiry date.

✓ System can prioritize earlier-expiring stock.

✓ CEO can view sales and performance remotely.

✓ Management can see daily profit and staff sales.

✓ Stock shortages and dead stock can be identified.

✓ Stock can be transferred between branches.

✓ Existing inventory can be imported from Excel.

✓ Business data can be exported whenever required.

✓ POS can operate during temporary internet failure.

✓ Data can be backed up and recovered.

✓ System can support multiple future Johncool Pharmacy branches.

FINAL DEVELOPMENT PRINCIPLE

This should be developed as a complete pharmacy management and business intelligence platform—not merely a cash register/POS. The system must give Johncool Pharmacy control from **STOCK** → **PURCHASE** → **BATCH** → **EXPIRY** → **SALES** → **PROFIT** → **STAFF** → **CASH** → **BRANCHES** → **MANAGEMENT**.

Prepared for: JOHNCOOL PHARMACY LIMITED

Document purpose: Software development specification and implementation reference.